

Feng Ye

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PRINCIPAL INTERESTS

My research interests span several areas in cyber-physical systems and distributed control systems, with a focus on distributed control and optimization, federated learning, smart grid, security and privacy preservation, attack detection, and their applications on microgrid, Internet of Vehicles, and Autonomous Underwater Vehicles.

APPOINTMENTS

Post-Doctoral Fellow Sept. 2024 - Present
University of Victoria, Victoria, BC, Canada

- Research in privacy preserving and security of federated learning, platooning control of Internet of Vehicles under direction of Prof. Lin Cai.

ACADEMIC BACKGROUND

Visiting Ph.D. Student, Electrical and Computer Engineering Oct. 2022 - Oct. 2023
University of Victoria, Victoria, BC, Canada

- Research in privacy preserving and security of federated learning under direction of Prof. Lin Cai.

Ph.D., Control Science and Engineering Sept. 2019 - Jun. 2024
Southeast University, Nanjing, Jiangsu, China

- Ph.D. research in privacy preserving and security of control systems under direction of Prof. Xianghui Cao and Prof. Mo-Yuen Chow.
- Dissertation title: Research on Privacy-Preserving Method for Distributed Economic Dispatch in Microgrids.

B.S., Electrical Engineering and Automation Sept. 2015 - Jun. 2019
Northeastern University, Shenyang, Liaoning, China

SPECIAL ACHIEVEMENTS

Selected Awards

- Southeast University Outstanding Ph.D. Dissertation Award, Southeast University, China, Oct. 2025 (Ranked 1st in the department, top 10% of Graduates)
- Best Technical Presentation Award, *Ubiquitous Intelligence Student Poster Competition*, IEEE Victoria Section, Canada, Dec. 2024
- People's Choice Award, *Workshop on Future Ubiquitous Networks 2023-Spring*, IEEE Victoria Section Joint VTS/ComSoc Chapter, Canada, May 2023
- Meritorious Winner Award, *Mathematical Contest in Modeling*, the Consortium for Mathematics and Its Applications, USA, Apr. 2018
- Honorable Mention Award, *Mathematical Contest in Modeling*, the Consortium for Mathematics and Its Applications, USA, Apr. 2017
- *Outstanding Student Cadres Award*, Northeastern University, Sept. 2017

Scholarships

- *Ph.D. Candidate 83791 Qiuzhen Scholarship*, Southeast University, China, 2024 (Competitive; 1/120 selected)
- *Ph.D. Candidate Scholarship*, Southeast University, China, 2020 - 2023
- *Postgraduate Student Scholarship*, Southeast University, China, 2019
- *National Encouragement Scholarship*, Ministry of Education, China, 2017
- *Undergraduate Student Scholarship*, Northeastern University, China, 2016 - 2019

Invited Lectures

- Privacy-Preserving Average Consensus and Its Application in Smart Grids, Central South University, 2024
- A Random-Weight Privacy-Preserving Algorithm With Error Compensation for Microgrid Distributed Energy Management, Workshop on Future Ubiquitous Networks 2023-Spring
- Simulating and Evaluating Privacy Issues in Distributed Microgrids: A Cyber-Physical Co-Simulation Platform, IECON 2021
- A Random-Weighted Privacy-Preserving Distributed Algorithm for Energy Management in Microgrid with Energy Storage Devices, IESSES 2020

RESEARCH EXPERIENCE

Byzantine-robust federated learning

Jan. 2023 - Sept. 2025

- Designed a Byzantine-robust federated learning algorithm to detect model poisoning attacks, which distinguishes malicious clients are based on probability distribution.

Secure control of distributed microgrids

Jan. 2021 - Dec. 2023

- Designed a collusion attack against distributed energy management systems based on the microgrid network topology, which undermines the economic utility of microgrids.
- Designed a novel false data injection attack detection method for distributed energy management systems based on the microgrid network topology, which can detect malicious devices in microgrids while preserve the data privacy of each device.

Privacy-preserving distributed coordinate control

Jan. 2020 - Jun. 2024

- Designed a novel privacy-preserving economic dispatch algorithm for distributed energy management system in microgrids, preserving privacy of the power information of devices.
- Drawed the upper bound of privacy-preserving performance of algorithms, and designed a multiplying-noise-based general privacy-preserving average consensus framework.

TEACHING EXPERIENCE

- *Guest Lecturer*, University of Victoria Fall 2024
ECE463/ECE514: Design and Analysis of Computer Networks
- *Guest Lecturer*, Southeast University Spring 2021
EE124: Large-Scale Grid Technology
- *Guest Lecturer*, Southeast University Spring 2020
SOA8122: System Identification and Modeling
- *Assistant and Lab Instructor*, Southeast University Spring 2020 - Spring 2024
SOA1180: Introduction to Communication Networks

STUDENT SUPERVISION

Undergraduate Thesis Supervision

- *Nianzhi Hang*, “A Simulation Platform for Secure Control in Distributed Smart Microgrid”, B.Eng. Thesis Co-Advisor under Prof. Xianghui Cao’s guidance., Southeast University, 2021
– Awarded Excellent Undergraduate Thesis of Southeast University

Undergraduate Project Supervision

- *Lengfeng Yu*, “Modeling, Control, and Optimization of F1 Tenth Cars”, Advisor, University of Victoria, 2025
- *Param Singh*, “Modeling, Control, and Optimization of F1 Tenth Cars”, Advisor, University of Victoria, 2025

PUBLICATIONS

Books

1. X. Cao, **F. Ye**, and K. Gui, *System Identification Theory and Practice*, China Machine Press, expected publication in 2026.

Journal Articles

2. [Auto25] **F. Ye**, X. Cao, L. Cai, and M.-Y. Chow, “False Noise Attack Detection for Differentially-Private Distributed Control of Microgrids,” *Automatica* (Regular Paper), vol. 179, 2025, Art. no. 112387.
3. [TIT24] **F. Ye**, X. Cao, M.-Y. Chow, and L. Cai, “Privacy-Preserving Average Consensus: Fundamental Analysis and a Generic Framework Design,” *IEEE Transactions on Information Theory*, vol. 70, no. 4, pp. 2870-2885, 2024.
4. [TSG23] **F. Ye**, X. Cao, Z. Cheng, and M.-Y. Chow, “CASL: A Novel Collusion Attack against Distributed Energy Management Systems,” *IEEE Transactions on Smart Grid*, vol. 14, no. 6, pp. 4717-4728, 2023.
5. [TIFS21] **F. Ye**, Z. Cheng, X. Cao, and M.-Y. Chow, “A Random-Weight Privacy-Preserving Algorithm With Error Compensation for Microgrid Distributed Energy Management,” *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 4352-4362, 2021.
6. [TSG21] Z. Cheng, **F. Ye**, X. Cao, and M.-Y. Chow, “A Homomorphic Encryption-Based Private Collaborative Distributed Energy Management System,” *IEEE Transactions on Smart Grid*, vol. 12, no. 6, pp. 5233-5243, 2021.

Conference Proceedings

7. [IESES20] **F. Ye**, Z. Cheng, X. Cao, and M.-Y. Chow, “A Random-Weighted Privacy-Preserving Distributed Algorithm for Energy Management in Microgrid with Energy Storage Devices,” *2020 2nd IEEE International Conference on Industrial Electronics for Sustainable Energy Systems (IESES)*, Cagliari, Italy, 2020, pp. 249-254.
8. [IECON21] N. Hang, **F. Ye**, Z. Cheng, X. Cao, and M.-Y. Chow, “Simulating and Evaluating Privacy Issues in Distributed Microgrids: A Cyber-Physical Co-Simulation Platform,” *47th Annual Conference of the IEEE Industrial Electronics Society(IECON)*, Toronto, ON, Canada, 2021.

Manuscripts under Review or in Preparation

9. [J1] **F. Ye**, L. Cai, and X. Cao, “Detection of False Data Injection Attacks in Remote State Estimation: A Parameter and Entropy-Based Approach,” *under review*.
10. [J2] **F. Ye**, P. Yang, L. Cai, and X. Cao, “TriGuardFL: Triple-Step Detection for Byzantine-Robust Federated Learning against Model Poisoning Attacks,” *under review*.

PROFESSIONAL SERVICE

Professional Activities

- Administrative Editor, *IEEE Transactions on Vehicle Technology*, Jan. 2026 - present
- Chair, *IEEE Young Professionals Victoria Section*, Jun. 2025 - present
- President, *UVic Chinese Ph.D. Students and Scholars Association*, Graduate Academics, Jun. 2025 - present
- Technical Program Committee member, *IEEE MASS 2025*
- Organization Committee member, *The Youth Scientist Workshop of Chinese Automation Association (Issue 19) and the 23rd Frontier Hot Topics for Automation*, Jan. 2022
- Organization Committee member, *The 14th “Going to Colleges and Universities” Series Workshop of the Youth Working Committee, Chinese Association of Automation and the 10th Control Theory and Application “Zhishan Frontier Forum” of Southeast University*, Nov. 2020
- Member, *IEEE*, 2024 - present
- Graduate Student Member, *IEEE*, 2020 - 2023

Peer Reviewer

- *Automatica*, 2022 - present
- *IEEE Transactions on Automatic Control*, 2022 - present
- *IEEE Transactions on Information Forensics and Security*, 2023 - present
- *IEEE Transactions on Industrial Electronics*, 2023 - present
- *IEEE Transactions on Industrial Informatics*, 2020 - present
- *IEEE Transactions on Control of Network Systems*, 2020 - present
- *IEEE Internet of Things Journal*, 2023 - present
- *IEEE Network Magazine*, 2023 - present
- *IEEE/CAA Journal of Automatica Sinica*, 2022 - present